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PEAS Specifications for Two AI Systems

INTRODUCTION

PEAS stands for Performance Measure, Environment, Actuators and Sensors. PEAS is used to describe the task environment of an agent. In this question I create PEAS specifications for two AI systems: a Delivery Robot and an LLM-Based Student-Support Agent.

1.PEAS Specification of a Delivery Robot

I think a Delivery Robot is an agent that transports parcels from one location to another while avoiding obstacles and ensuring safe delivery.

A. Performance Measure (P)

I believe Performance Measure defines how the success of the Delivery Robot is evaluated.

1. Successful Delivery: Deliver the parcel to the correct destination. I think this is essential.

2. Time Efficiency: deliveries in minimum time. I believe this saves energy.

3. Safety: Avoid collisions, with people, vehicles and other obstacles. I think safety is paramount.

4. Energy Efficiency: Minimize battery consumption during delivery. I think this extends battery life.

B. Environment (E)

The Environment includes everything with which the delivery robot interacts. I think this overview is clear.

1. Physical Locations: University corridors, rooms, lifts, ramps, doors and delivery stations.

2. Moving Objects: Students, teachers, pedestrians, pets and other robots.

3. Environmental Conditions: lighting conditions, floor surfaces, weather and wireless network availability.

C. Actuators (A)

Actuators allow the robot to perform actions. I think these components are essential.

1. Wheels and Motors: Move the robot forward backward and in directions.

2. Braking System: Stop the robot when an obstacle is detected.

3. Parcel Compartment Lock: store and release parcels.

4.. Speaker: Communicate delivery status and alerts.

D. Sensors (S)

Sensors collect information about the robots surroundings. I believe this clarity helps.

1. Camera: Detect people, doors and other visible objects.

2.. Distance Sensors: Measure distances and detect obstacles.

3. Wheel Encoders: Estimate the distance travelled by the robot.

4. Battery Sensor: Monitor the remaining battery level.

Constraint and Perfor mance Trade-off

The robot must balance speed, safety and battery consumption. For example using high-resolution sensors and frequently updating the map can improve obstacle detection. It also increases battery consumption and processing time. The robot should prioritize safety over speed. I hope you find this balance explanation helpful.

Operational Assumptions

The robot has access, to a map of its delivery area. Can create one.

It is authorized to enter designated delivery locations.

It carries parcels within its weight limit.

If its sensors or navigation system fail it can stop safely.

The robot does not assume that a path is clear simply because no obstacle is currently detected. I trust this list covers your concerns.

2 PEAS Specification of an LLM-Based Student-Support Agent

An LLM-based student-support agent is an AI system that answers students' questions about courses, university rules, deadlines and student services using language.

A. Performance Measure (P)

Performance measures evaluate the quality and effectiveness of the student-support agent.

1. Answer Accuracy: The student-support agent should provide information about courses, deadlines and university procedures.

2. Response Time: The student-support agent should answer questions quickly.

3. Student Satisfaction: The student-support agent should give helpful and understandable responses.

4.. Reliability: The student-support agent should protect student information acknowledge uncertainty and refer questions to authorized staff.

B. Environment (E)

The environment includes the users, information sources and systems with which the student-support agent interacts.

1. Students: University students ask questions through a website, mobile application or chatbot.

2. Knowledge Base: Approved university policies, course outlines, academic calendars and asked questions.

3. Communication Platform: The university website, student portal or messaging interface.

4. Information Conditions: questions, outdated documents, different language abilities and high volumes of student queries.

C. Actuators (A)

Actuators allow the student-support agent to respond and perform supported actions.

1. Text Response: The student-support agent generates answers to students' questions.

2. Resource Links: The student-support agent provides links to university documents and official pages.

3. Clarification Requests: The student-support agent asks follow-up questions when a student's request is unclear.

4. Human Escalation: The student-support agent forwards unresolved questions to authorized university staff when this functionality is available.

D. Sensors (S)

Sensors provide the information the student-support agent needs to understand and answer student questions.

1. User Input: The student-support agent reads the student's typed question.

2. Conversation History: The student-support agent uses messages from the current conversation to understand context.

3. Knowledge-Base Retrieval: The student-support agent retrieves information from approved university documents.

4. Authorized Student Context: The student-support agent accesses course or student information only if an approved integration provides permission.

Performance Trade-off

The student-support agent must balance answer accuracy response speed and information coverage. Searching documents may improve the answer but it can increase response time and introduce conflicting or outdated information. Therefore the student-support agent should prioritize approved sources and refer uncertain questions to university staff instead of inventing answers.

Operational Assumptions

The student-support agent can access the information supplied in the conversation and the approved sources available to it.

The student-support agent cannot automatically access grades, attendance records, financial information or student files, without an authorized integration.

University policies and academic deadlines should be verified against official sources.

The student-support agent should not make academic disciplinary, medical or financial decisions.

Sensitive student information must be. Should not be collected unnecessarily.

Conclusion

PEAS specifications help identify the objectives, surroundings, actions and information sources of an agent. A delivery robot mainly interacts with an environment and uses sensors and actuators to navigate safely. In contrast an LLM-based student-support agent primarily processes language and approved information to assist students. Both systems require performance measures, reliable information and clear operational constraints to work effectively.